10 Nuclear power stations use uranium as the fuel.

(a) An isotope of uranium has 92 protons and 143 neutrons in each nucleus.

Complete the nuclide notation for this isotope.

.....
U
.....

[2]

(b) Complete the following sentences:

In a nuclear reactor, a slow-moving ..... hits a uranium nucleus.

This causes the uranium nucleus to split into smaller ..... .

This reaction is called nuclear ..... .

The reaction releases a large amount of energy.

This energy heats ..... to drive a turbine which turns a

..... to produce electricity.

[3]

(c) Step-up transformers produce high-voltage electricity that is transmitted from the power station to consumers far away from the power station.

(i) Explain why the power loss in the cables during electricity transmission is smaller when the transmission is at a high voltage.

.....

.....

..... [2]

(ii) A transformer in the power station steps up the voltage from 25 000 V to 400 000 V. The transformer has 240 turns on the primary coil.

Calculate the number of turns on the secondary coil of the transformer.

number of turns = ... [2]

[Total: 9]

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